Dr. Vanessa R. MARCELINO

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Google Scholar Link

CURRENT POSITIONS

2025-pres	**Ramon y Cajal Researcher.** Instituto de Agroquímica y Tecnología de Alimentos (IATA – CSIC), Valencia, Spain.
2025-pres	**Honorary Research Fellow.** Peter Doherty Institute for Infection and Immunity, University of Melbourne, Australia.

PREVIOUS POSITIONS

2023-2025	**Group Leader, ARC DECRA Fellow.** Melbourne Integrative Genomics and Peter Doherty Institute for Infection and Immunity, University of Melbourne, Australia
2020-2023	**Research Fellow in Genomics and Bioinformatics.** Hudson Institute of Medical Research, Monash University, Australia. Advisor: Samuel Forster.
2017-2019	**Research Fellow in Pathogen Genomics.** Westmead Institute for Medical Research, University of Sydney, Australia. Advisors: Tania Sorrell and Edward C. Holmes.

EDUCATION AND KEY QUALIFICATIONS

2017	**PhD**	School of BioSciences, University of Melbourne, Australia Supervisor: Heroen Verbruggen
2012	**MSc**	Erasmus Mundus MSc in Marine Biodiversity, Bremen U. (Germany); U. Pierre et Marie Curie (France); Ghent U. (Belgium)
2010	**Licen.**	Biological Sciences, São Paulo State University, Brazil
2010	**BSc**	Biological Sciences, São Paulo State University, Brazil

CV SUMMARY

I work at the interface between data science and microbiome ecology. I develop and apply bioinformatic approaches to understand microbial communities that influence food safety, human health and natural ecosystems.

Microbial communities play an important role in the health, fitness and productivity of the host or environment where they live in, but we do not fully understand how. We have estimates of which microbial species exist and where to find them, but we have little knowledge about what these species are doing in those communities, how they interact with each other, and how they influence their host or environment. Obtaining a better understanding of such complex systems is a major data science challenge, the domain where I have made most of my contributions. My research program focuses on three core areas that are directly applicable in the Food Science and Technology field, including: **1)** development of bioinformatic tools for microbial identification and pathogen detection (e.g. Marcelino et al. *Genome Biology* 2020; Clausen et al. *NAR Genomics and Bioinformatics* 2025), which are applied to monitor pathogens in agrifood systems; **2)** leveraging computational modelling of microbial interactions to propose targeted microbiome engineering strategies (e.g. Marcelino et al. *Nature Communications* 2023), setting foundations for precision nutrition and the development of functional foods; and **3)** characterising antimicrobial resistance in microbiomes of humans, livestock and wild animals (e.g. Marcelino et al *BMC Biology*; Shi et al *CELL*), contributing data that has already been used by the World Health Organisation.

My work has attracted **over 2000 citations** (Google Scholar), and my **h-index is 25**, which is high for my career stage (<9 years post-PhD). I have 44 peer-reviewed publications. These include 15 publications as first author (of which 13 I am both first and corresponding author), and 9 as second author, highlighting my leading contribution in these studies. The majority of my papers (79%) were published in Q1 journals, including in high-impact journals such as *CELL*, *Nature Communications*, *Nature Methods, Current Biology*, *Nature Microbiology*, *ISME J*, *Immunity*, *Microbiome,* and *Cell Reports Medicine*, demonstrating my ability to produce high-quality scientific outputs.

In 2023 I was hired as an **independent group leader at the University of Melbourne** – ranked the best university in Australia (Times Higher Education and QS rankings). In June 2025 I moved to Spain to join **IATA-CSIC, supported by a Ramon y Cajal Fellowship**. I formed collaborations with 3 different groups at IATA within the short time that I have been there, which have already led to two successful grant applications and a manuscript under review. Beyond IATA, I have an extensive network of collaborations that has resulted in publications with authors from 18 countries, and I have well-established links with world-leading groups in Denmark, Australia and Germany. My research program is supported by **>4M EUR in competitive funding that I have earned as lead or co-investigator**, in addition to several international projects where I contribute bioinformatics expertise as a team member, including a recently funded HORIZON project.

GRANTS, AWARDS AND FELLOWSHIPS

Foreign currencies converted to Euros using Banco de España's "conversor de divisas". Based on the 01st of February of the first year of the grant.

>> As sole PI [€770k EUR]

1. Atracción de talento – CSIC (2025 – 2028). €100.000. *Characterising plasticity in the human gut microbiome.*

2. Ayudas para contratos Ramón y Cajal (2025 – 2030). €120.000 + salary. Grant ID: RYC2023-042907-I.

3. ARC Discovery Early Career Researcher Award (2022 – 2025). €423.618 (ca. €90k for research, the remaining for salary). *Understanding nutritional interactions for targeted microbiome manipulation*. Grant ID: DE220100965.

4. Tony Basten Award, University of Sydney (2019). €38.001. *Antimicrobial treatment effects on fungal-bacterial interactions.*

5. Microbial Eco-Evolutionary Dynamics Symposium travel support (2018). €500.

6. Postgraduate Writing-Up Award (2017). €2.108

7. Holsworth Wildlife Research Endowment (2013 – 2017). €11.422. *Biodiversity and evolution of limestone-boring algae – insights from the Great Barrier Reef.*

8. Sophie Ducker Postgraduate Scholarship (2016). €1.431 EUR

9. Australian Biological Resources Study Travel Awards (2013, 2015 and 2016). €2.132

10. School of Botany Travel Award (2014). €1.142

11. The David Ashton Travel Award (2013). €990

12. PhD scholarship: Melbourne International Research Scholarship (2013 – 2016). €18.069 per year, + university fee remission.

13. MSc scholarship: Erasmus Mundus Master of Science in Marine Biodiversity and Conservation (2010 – 2012). €23.500 per year, + university fee remission.

14. Technical Training Scholarship (2010). €8.162

15. Scientific Initiation Scholarship (2006 – 2009). €6.735 per year.

>> Collaborative grants as co-investigator (co-PI) [€3,4M EUR]

16. IMPULSA 2026, Severo Ochoa, IATA-CSIC (2026, co-investigator in a team of 2). €30.000. *Unravelling diet-driven microbiome dynamics using tailored synthetic communities*. [PIs: MS Royo – *Bacterias Lácticas y Probioticos* group, and VR Marcelino – *GutBioTox* Group]. This is a **newly established, cross-departmental collaboration within IATA.** I co-designed and co-wrote the proposal, and lead the bioinformatics and computational modelling work.

17. NHMRC Synergy Grant (2025 – 2029, co-investigator in a team of 6). **€2.993.653.** *Exploiting interactions between microbiota and T cells to improve melanoma immunotherapy.* [PI: Sandhu, Grant ID: 2036149]. I am the microbiome expert in the team. I contributed to grant conceptualization and writing. **I supervise a postdoc and I continue to lead the microbiome data analyses components.**

18. MDHS EMCR Project Catalyst Grants (2024 – 2025, co-investigator in a team of 3). €22.926. *Investigating the production, engraftment, and impact of faecal microbiome transplantation*. [PI: Welsh]. I contributed to writing and advised the metabolic modelling components of the project.

19. Cumming Global Centre for Pandemic Therapeutics Foundation Grants (2024 – 2026, co-investigator in a team of 10). €364.342. *Customisable biotherapeutics against pandemic pathogens*. [PI: Howden]. I contributed to project design. My role is to guide the development of computational resources to speed-up the design of microbiome-based interventions.

20. DMI Collaborative Grants, Doherty Institute (2023, co-investigator in a team of 2). € 12.923. *Defining how commensal gut bacteria can kill multidrug resistant pathogens.* [PI: Carter]. I helped design the project and provided metabolic modelling expertise. This work formed the basis of the project later funded by the Cumming Centre (#19).

21. Sydney Institute of Agriculture (2019, co-investigator in a team of 12). €63.335. *Agricultural DMIs as drivers of anti-fungal resistance in plant, animal, and human pathogens: a One Health investigation.* [PI: Beardsley]. I advised the genomics and bioinformatics components of the project.

22. Marie Bashir Institute for Infectious Diseases (2018-2019, **I was the lead investigator** in a team of 5). €12.847. *The human gut mycobiome responses to antimicrobial treatment*. I conceptualised and wrote the project, and led the experimental and computational analyses of fungal genomes.

>> Collaborative grants as associate investigator or named team member [€5,4M EUR]

23.Horizon Europe (2026 – 2029, team member). **€3.999.999**. FLOWSAFE: *Restoration of FLOod-degraded ecosystems With Synergistic, tAilored bio-/nature-based remediation oF pollutant mixturEs.* [PI: M. Arroyo, Grant ID: 101291038]. **I co-designed the microbiome data analysis strategy in close collaboration with the project coordinator at IATA** (Marta Arroyo – *GutBioTox* group). I will supervise a PhD student working on this project.

24. NHMRC Ideas Grants (2021 – 2025, associate investigator). €887.614. *Should you be eating that? Food-derived bacteria and their role in treating disease*. [PI: Forster, Grant ID: 2000701]. **I performed the preliminary data analysis of food-borne microbiomes that formed the basis of the proposal.** I helped writing the proposal. I supervised a student working on the project, in addition to advise the data analyses and community ecology aspects.

25. NHMRC Ideas Grants (2021 – 2024, associate investigator). €532.555. *Interferon Epsilon as a novel regulator of the host-microbiome response in the pathogenesis of inflammatory bowel disease*. [PI: Giles, Grant ID: 2003918]. I performed the multi-omics data integration analysis that generated key hypotheses for the proposal. I advised the microbiome components of the project.

26. NHMRC Centre for Research Excellence in Tuberculosis Control (2021, associate investigator). €12.633. *Impact of MDR-LTBI therapy on bacterial antimicrobial resistance profiles and microbiome perturbations.* [PI: Howard-Jones]. I helped conceptualise the project, assisted the project from sample collection to sequencing and bioinformatics, performed the microbiome data analyses and significantly contributed to writing the resulting manuscript (Howard-Jones, Marcelino, et al. currently under review at Lancet Microbe).

27. NSW Public Health Pathogen Genomics Partnership (2019, associate investigator). €9.500. *Assessing the zoonotic potential of Candida spp.* [PI: Beardsley]. I was the bioinformatics and genomics expert in the team.

SUPERVISION AND MENTORING

2024-pres	K Tandon, Postdoctoral Researcher. **Principal supervisor**. University of Melbourne
2024-pres	Z Wu, PhD student. **Principal supervisor**. University of Melbourne
205-pres	AM Garcia, MSc student. **Co-supervisor,** University of Valencia
2024-pres	Y Tong. **Co-supervisor**, University of Melbourne
2024-2025	M Spralja, MSc student. **Co-supervisor,** University of Melbourne
2021-2025	V Salazar (PhD student). **Co-supervisor**, University of Melbourne
2024	M Hallgren, PhD student (**Principal supervisor during internship**), Tech. Univ. Denmark
2020-2024	R Young (PhD student). **Co-supervisor**, Monash University
2020-2022	S Solari (intern and Honours student). **Mentor**, Monash University
2021	R Sundararajan, (research assistant). **Principal supervisor**, Hudson Institute
2019	K Kim, (research assistant). **Sole supervisor**, Westmead Institute, Sydney
2019	P Clausen, PhD student (**Principal supervisor during internship**), Tech. Univ. Denmark
2017-2018	C Cremen, PhD student. **Co-supervisor**, University of Melbourne

TEACHING AND OTHER INSTITUTIONAL RESPONSIBILITIES

2023-2024	**Academic Mentor**. 9 undergraduate students. University of Melbourne
2023-2024	**Academic Mentor**. Genetic Analysis course – Journal Club. University of Melbourne
2023	**Guest lecturer**. Medical Microbiology. University of Melbourne
2017	**Guest lecturer and Demonstrator**. Genomics and Bioinformatics. University of Melbourne
2016-2017	**Demonstrator**. Blue Planet. University of Melbourne
2013-2016	**Demonstrator**. Marine Botany. University of Melbourne
2013-2017	**Demonstrator**. Biology of Cells and Organisms. University of Melbourne
2014	**Teaching Assistant**. Phylogenetics workshop associated with Brazilian Phycological Congress
2009	**Teaching Assistant.** Genetics. Universidade Estadual Paulista (Brazil).
2009&2012	**Elementary and high school teacher.** Biology. (Brazil)
2005-2006	**Educator.** Environmental awareness. Rio Claro City Council, Brazil.
2023-2025	**Communications Committee.** Melbourne Integrative Genomics, University of Melbourne
2020-2023	**Bioinformatics Manager**. Australian Microbiome Culture Collection, Hudson Institute

DIVERSITY AND INCLUSION

I am committed to promote an inclusive and equitable research group. Diversity and inclusion are part of the *Appropriate Workplace Behaviour* training that I completed while at the University of Melbourne. These topics are also covered during the course required to be a PhD supervisor at the University of Melbourne. The diversity in my past and current team is remarkable: I supervised 8 males, 6 females, from 8 different countries. I am a first-generation college graduate myself. I am aware that I am not immune to unconscious biases, and actively reflect on them to ensure fair evaluation, supervision, and collaboration practices. I am committed to promote equality, fairness and transparency, supported for example by IATA's Comité de Igualdad.

PRIZES

Best short oral presentation at the Hospital Week Research Symposium, Sydney, Australia (2018).

Honourable mention. 14th Environmental Microbiology Meeting, João Pessoa, Brazil (2014).

Best poster. 54th Brazilian Congress of Genetics. Salvador, BA, Brazil (2008).

Best poster. 3rd Brazilian Congress of Herpetology. Belém, PA, Brazil (2007).

As supervisor, I have mentored my students in designing award-winning presentations at national (4×) and international (2×) conferences.

ORGANISATION OF SCIENTIFIC MEETINGS

2024 **Organising Committee:** BioInfoSummer Workshop, Melbourne.

2022 **Organising Committee:** Australian Microbial Ecology Meeting, Melbourne.

2019 **Organising Committee:** Yeasts: Products and Discovery. Sydney.

2010 **Organising Committee:** Anuran Taxonomy Symposium. Rio Claro, Brazil.

2022 **Session Organiser and Chair:** Human microbiome. Australian Microbial Ecology meeting.

2019 **Session Organiser and Chair:** Microbiome and Biofilms. Yeasts: Products and Discovery.

2019 **Session Chair:** Microbiome in Health and Disease. Australian Society for Microbiology.

INVITED TALKS AND SELECTED MEETINGS

2024 **Invited speaker:** Microbial interactions and human health. *Australian Society of Microbiology (ASM) meeting*, Brisbane, Australia.

2024 **Invited speaker:** Charting metabolic interactions between gut microbes. *Australian Bioinformatics And Computational Biology Society (ABACBS) National Seminar Series*. Online, Australia.

2023 **Plenary speaker:** Unravelling microbiomes. *Metagenomics workshop*, Copenhagen, Denmark.

2022 **Invited speaker:** Microbiome classification from metagenome data. *BioInfoSummer*, Melbourne.

2021 **Plenary speaker:** A quest for accurate microbiome classification from metagenome data. *MicroSeq* (*Austr. Soc. Microbiol*.), online, Australia.

2021 **Invited speaker:** Metagenomics – applications and challenges to study microbial communities. *Python for Biologists workshop*. Online, Brazil.

2021 **Invited speaker:** Accurate microbiome classification from metagenome data. *Microbial Ecology-Environmental Microbiology meeting*, Melbourne, Australia.

2019 **Invited participant:** Ideas Lab (micro-eukaryote 'omics, NSF RCN EukHiTS), UC San Diego, USA.

2016 **Invited participant & speaker:** Into the Genome. Residential meeting of The Royal Society, London, Buckinghamshire, UK.

2025 **Guest speaker.** *FISABIO*, Valencia, Spain.

2023 **Guest speaker.** *Instituto Gulbenkian de Ciência*, Lisbon, Portugal.

2023 **Guest speaker.** *Globe Institute*, Copenhagen, Denmark.

2023 **Guest speaker.** *BIOPOLIS/CIBIO*, Porto, Portugal.

2022 **Guest speaker.** *Australian National Univ.*, Canberra

2021 **Guest speaker***. Univ. of New South Wales*, Sydney, Australia.

Selected oral presentations (from over 30 conference presentations):

2024 VR Marcelino et al*. Disease-specific loss of microbial cross-feeding interactions in the human gut.* **19th International Symposium on Microbial Ecology**, Cape Town, South Africa.

2021 VR Marcelino et al. *AusMiCC – A New Australian Microbiome Culture Collection.* **Australian Society for Microbiology Meeting**. Online, Australia.

2019 VR Marcelino. *No more discrimination – including microbial eukaryotes in metagenomic surveys*. **Society for Molecular Biology and Evolution Meeting**. Manchester, UK.

2018 VR Marcelino et al. *Antibiotic resistance in the microbiome of wild birds*. **Sydney Bioinformatics Research Symposium**. Sydney, Australia.

2015 VR Marcelino and H Verbruggen. *Diversity and evolutionary origins of microalgae burrowing in coral skeletons revealed through environmental sequencing*. **Evolution 2015** – joint meeting of Soc. for the Study of Evolution, American Soci. of Naturalists and the Soc. of Systematic Biologists. Guarujá, Brazil.

COMMISSIONS OF TRUST

2022-pres **Grant Assessor**, Australian Research Council (10×)

2019 **Grant Assessor**, Royal Melbourne Hospital (1×)

2026 **PhD examination board member:** DTU Food Department, Technical University of Denmark.

2021 **MSc thesis reviewer:** Molecular Sciences Department, Macquarie University

2018-pres **Honours theses reviewer** (2× for U. Melbourne/Australia and 1× for U. Uberlandia/Brazil).

2016-pres **Peer reviewer** for >20 journals including: *CELL, Nature Microbiology*, *Bioinformatics Advances*, Molecular Biology and Evolution, *eLife*, *PCI Genomics*, *Genome Biology*, *Molecular Ecology*, *Integrative and Comparative Biology*, *Trends in Microbiology*.

RESEARCH TRANSLATION MERITS

1. Pathogen detection in farmed products. I have provided advice for industry and government bodies using the software I developed (CCMetagen) to detect pathogens in agricultural products, including: PlantTech Research Institute (horticulture, New Zealand), Pathosense (veterinary medicine, Belgium), Department of Primary Industries of New South Wales (beef cattle, Australia), and the Pacific Northwest (PNW) Research Station (US Forest Service).

2. Helping the World Health Organization (WHO) identify the risks associated with preventive antibiotics for Tuberculosis (TB). It has become common to prescribe antibiotics to close contacts of individuals with TB, even when they do not have symptoms (known as cases of latent TB). This practice is controversial due to the rise of antimicrobial resistance and the potential consequences for gut health. With collaborators at the University of Sydney, I obtained seed funds for metagenome sequencing of samples from the latest clinical trial to evaluate the effectiveness of preventive antibiotic treatment in individuals with latent TB [Project "*Impact of MDR-LTBI therapy on bacterial antimicrobial resistance profiles and microbiome perturbations*", $20,000 AUD, PI: A. Howard-Jones, my role: associate investigator]. I designed a computational pipeline and performed the bioinformatic analyses of the metagenome data. Our results were presented and discussed at the meeting of the organisation that defines and revises the WHO's endorsed practices (WHO's Guideline Development Group on Latent Tuberculosis meeting, December 2023).

CAREER BREAK

2020-2021 Work interruptions due to strict lockdowns (lasting 263 days in Melbourne) led to the postponement (1×) or cancelation (2×) of the main projects of my postdoctoral work.

2021-2023 Maternity leave: equivalent of 14 months interruption. Breakdown: October 2021 to April 2022 at 1 Full Time Equivalent (FTE); May 2022 to January 2023 at 0.5 FTE; Feb 2023 to 14th March 2024 at 0.2 FTE.

PUBLICATIONS

Total citations = 2091, h-index = 25 (Google Scholar, Jan 2026)
An asterisk (*****) indicates publications where I am corresponding author.

Preprints:

47. M Fernández-Niño, LA Wessjohann, JM Guillamón, D Burgos-Toro, S Moriano-Gutierrez, G Di Matteo, Z Islam, R van Tatenhove-Pel, **VR Marcelino**, R Ledesma-Amaro. From circuits to symphonies: redefining synthetic biology for multimicrobial systems. *preprint* 2025. DOI: 10.20944/preprints202509.1416.v1

46. Y Tong, **VR Marcelino**, R Turnbull, H Verbruggen. ChloroScan: Recovering plastid genome bins from metagenomic data. *arXiv* 2025; 2510.10950. DOI: 10.48550/arXiv.2510.10950

45. VW Salazar, H Verbruggen, **VR Marcelino**[#], K-A Lê Cao[#] ([#]co-senior authors). Global picoplankton biogeography revealed by metagenomic and climatic data integration. *bioRxiv* 2024; 2024-11. DOI: 10.1101/2024.11.23.624595

Peer-reviewed publications:

44. Y Shi[1], Y Li[1], H Li[1], A Haerheng[1], **VR Marcelino**[1], M Lu, P Lemey[1], J Tang, Y Bi, JH Pettersson[1], J Bohlin[1], J Klaps, Z Wu, W Wan, B Sun, M Kang, EC Holmes[1], N He, S Su. Extensive cross-species transmission of pathogens and antibiotic resistance genes in mammals neglected by public health surveillance. *CELL* 2025; 23, 6591 – 6605.e14. ([1] equal contribution).

43. C Welsh, PR Cabotaje, **VR Marcelino**, TD Watts, DJ Kountz, M Jespersen, JA Gould, NQ Doan, JP Lingford, T Koralegedara, J Solari, GL D'Adamo, P Huang, N Bong, EL Gulliver, RB Young, H Land, K Walter, I Cann, GV Pereira, EC Martens, PG Wolf, JM Ridlon, HR Gaskins, EM Giles, D Lyras, R Lappan, G Berggren, SC Forster, C Greening. A widespread hydrogenase drives fermentative growth of gut bacteria in healthy people. *Nature Microbiology* 2025; 10, 2686–2701

42. A Bachem, M Clarke, G Kong, I Tarasova, L Dryburgh, L Kosack, AR Lee, LD Newland, T Wagner, E Bawden, KM Yap, MD Wilson, S Engel, KR Wilson, BE Russ, K Tandon, PKH Lau, G McArthur, **VR Marcelino**, PA Beavis, SJ Turner, J Waithman, TP Stinear, S Sandhu, J Schröder, T Gebhardt, S Bedoui.

SCFA producing capabilities of the gut microbiota enhance tumor-specific CD127+CD8+ T cell immunity against melanoma. *Immunity* 2025; 58(11) 2799-2813.

41. PT Clausen, HB Hallgren, S Overballe-Petersen, **VR Marcelino**, H Hasman, & Aarestrup, F. M. Assembly-free typing of Nanopore and Illumina data through proximity scoring with KMA. *NAR Genomics and Bioinformatics* 2025; 7(3), lqaf116.

40. H Koceva, M Amiratashani, P Akbarimoghaddam, B Hoffmann, G Zhurgenbayeva, MS Gresnigt, **VR Marcelino**, C Eggeling, MT Figge, M-J Amorim, AS Mosig. Deciphering respiratory viral infections by harnessing organ-on-chip technology to explore the gut–lung axis. *Open biology* 2025, 15(3), 240231.

39. A Day, R Slater, **VR Marcelino**, R Wheeler, R Young, N Maddigan, S Forster, S Costello, W Uylaki, C Probert, J Andrews, CK Yao, P Gibson, R Bryant. Deep functional analysis of a defined sulphide-reducing diet (4-SURE diet) as a novel therapy for ulcerative colitis (UC) demonstrates all specific luminal micro environmental objectives were achieved. *Inflammatory bowel diseases* 2025; 31, no. 11 (2025): 3160-3171.

38. LK Boelsen, MJ Williams, YTM Mangwiro, H Hansji, A Czajko, **VR Marcelino**, S Forster, JM Said, C Satzke, R Saffery. The potential of residual clinical Group B *Streptococcus* swabs for assessing the vaginorectal microbiome in late pregnancy. *Scientific Reports* 2024; 14(1), 19318.

37. VR Marcelino*. Gut health: the value of connections. *eLife.* 2023; 12, e92319.

36. VR Marcelino*, C Welsh, C Diener, EL Gulliver, EL Rutten, RB Young, EM Giles, SM Gibbons, C Greening, SC Forster. Disease-specific loss of microbial cross-feeding interactions in the human gut. *Nature Communications* 2023; 14, 6546.

35. VW Salazar, B Shabban, MM Quiroga, R Turnbull, E Tescari, **VR Marcelino**, H Verbruggen, K-A Lê Cao. Metaphor – A workflow for streamlined assembly and binning of metagenomes. *GigaScience.* 2023; 12, giad055.

34. VR Marcelino*, C Birnbaum. AusME 2022 Melbourne conference. *Microbiology Australia* 2023; 44, 3-4.

33. SA Marshall, RB Young, JM Lewis, EL Rutten, J Gould, CK Barlow, C Giogha, **VR Marcelino**, N Fields, RB Schittenhelm, EL Hartland, NE Scott, SC Forster, EL Gulliver. The broccoli-derived antioxidant sulforaphane changes the growth of gastrointestinal microbiota, allowing for the production of anti-inflammatory metabolites. *Journal of Functional Foods* 2023; 107, 105645.

32. GL D'Adamo, M Chonwerawong, LJ Gearing, **VR Marcelino**, JA Gould, EL Rutten, SM Solari, PWR Khoo, TJ Wilson, T Thomason, EL Gulliver, PJ Hertzog, EM Giles, SC Forster. Bacterial clade-specific analysis identifies distinct epithelial responses in inflammatory bowel disease. *Cell Reports Medicine* 2023; 4:101124.

31. EL Gulliver, V Adams, **VR Marcelino**, J Gould, EL Rutten, DR Powell, RB Young, GL D'Adamo, J Hemphill, SM Solari, SA Revitt-Mills, S Munn, T Jirapanjawat, C Greening, JC Boer, KL Flanagan, M Kaldhusdal, M Plebanski, KB Gibney, RJ Moore, JI Rood, SC Forster. Extensive genome analysis identifies novel plasmid families in *Clostridium perfringens*. *Microbial Genomics* 2023; 9: mgen000995.

30. SM Solari, RB Young, **VR Marcelino**, SC Forster. expam—high-resolution analysis of metagenomes using distance trees. *Bioinformatics* 2022; btac591.

29. EL Gulliver, RB Young, M Chonwerawong, GL D'Adamo, T Thomason, JT Widdop, EL Rutten, **VR Marcelino,** RV Bryant, SP Costello, CL O'Brien, GL Hold, EM Giles, SC Forster. The future of microbiome-based therapeutics. *Alimentary Pharmacology & Therapeutics* 2022; 00: 1-17.

28. MM Pasella, M-FE Lee, **VR Marcelino**, A Willis, H Verbruggen. Ten *Ostreobium* (Ulvophyceae) strains from Great Barrier Reef corals as a resource for algal endolith biology and genomics. *Phycologia* 2022; 4:452-458

27. F Meyer, A Fritz, Z-L Deng et al. Critical Assessment of Metagenome Interpretation – the second round of challenges. *Nature Methods* 2022; 19: 429-440

26. ACM Teixeira, **VR Marcelino**, J Alexandrino, CFB Haddad, AA Giaretta. Populational differentiation in *Boana bischoffi* (Anura, Hylidae): revisiting the issue using molecular, morphological, and acoustic data. *Journal of Herpetology* 2022; 56: 110-119

25. T Jia, WS Chang, **VR Marcelino**, S Zhao, X Liu, Y You, EC Holmes, M Shi, C Zhang. Characterization of the gut microbiome and resistomes of wild and zoo-captive macaques. *Frontiers in Veterinary Science* 2022; 8:778556.

24*.* R Young, **VR Marcelino,** M Chonwerawong, EL Gulliver, SC Forster. Key Technologies for Progressing Discovery of Microbiome-Based Medicines. *Frontiers in Microbiology* 2021; 12:1604

23. B Jones-Freeman B, M Chonwerawong, **VR Marcelino**, AV Deshpande, SC Forster, MR Starkey. The microbiome and host mucosal interactions in urinary tract diseases. *Mucosal Immunology* 2021; 4:1-4.

22. C Iha, JA Varela, V Avila, CJ Jackson, KA Bogaert, Y Chen, LM Judd, RR Wick, KE Holt, MM Pasella, F Ricci, SI Repetti, M Medina, **VR Marcelino**, CX Chan, H Verbruggen. Genomic adaptations to an endolithic lifestyle in the coral-associated alga *Ostreobium*. *Current Biology* 2021; 31:1393-1402

21. J Charon, **VR Marcelino**, R Wetherbee, H Verbruggen, EC Holmes. Meta-transcriptomic detection of diverse and divergent RNA viruses in green and chlorarachniophyte algae cultures. *Viruses* 2020; 12:1180

20. VR Marcelino*, EC Holmes, TC. Sorrell. The use of taxon-specific reference databases compromises metagenomic classification. *BMC Genomics* 2020; 21:184

19. VR Marcelino*, PTLC Clausen, JP Buchmann, M Wille, JR Iredell, W Meyer, O Lund, TC Sorrell, EC Holmes. CCMetagen: comprehensive and accurate identification of eukaryotes and prokaryotes in metagenomic data. *Genome Biology* 2020; 21:103.

18. A Góes-Neto, **VR Marcelino**, H Verbruggen, FF da Silva and F Badotti. Biodiversity of endolithic fungi in coral skeletons and other reef substrates revealed with 18S rDNA metabarcoding. *Coral Reefs* 2020; 39:229

17. F Ricci, **VR Marcelino**, LL Blackall, M Kühl, M Medina, H Verbruggen. Beneath the surface: community assembly and functions of the coral skeleton microbiome. *Microbiome* 2019; 7:159.

16. VR Marcelino*, L Irinyi, JS Eden, W Meyer, EC Holmes, TC Sorrell. Metatranscriptomics as a tool to identify fungal species and subspecies in mixed communities – a proof of concept under laboratory conditions. *IMA Fungus* 2019; 10:8

15. R Wetherbee, **VR Marcelino**, JF Costa, B Grant, S Crawford, RF Waller, RA Andersen, D Berry, GI McFadden, H Verbruggen. A new marine prasinophyte genus alternates between a flagellate and a dominant benthic stage with microrhizoids for adhesion. *Journal of Phycology* 2019; 55:1210-1225

14. VR Marcelino*, M Wille, AC Hurt, D González-Acuña, M Klaassen, JS Eden, M Shi, JR Iredell, TC Sorrell, EC Holmes. Meta-transcriptomics reveals a diverse antibiotic resistance gene pool in avian microbiomes. *BMC Biology* 2019; 17:31.

13. C Biswas, **VR Marcelino,** S Van Hal, C Halliday, E Martinez, Q Wang, S Kidd, K Kennedy, D Marriott, CO Morrissey, I Arthur, K Weeks, MA Slavin, TC Sorrell, V Sintchenko, W Meyer, SC-A Chen. Whole genome sequencing of Australian *Candida glabrata* isolates reveals genetic diversity and novel sequence types. *Frontiers in Microbiology* 2018; 9:2946

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